MediBridge

Turning Hindi–English (Hinglish) clinical notes into readable, structured English records

NLP Project · Industry Track · Milestone M1 — Problem Definition & Business Case (revised with M2 data findings)

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One OPD note: two languages, no structure

The note as the doctor writes it

“Patient ko 3 din se tez bukhar hai, sar dard bhi hai. Paracetamol 650 mg din mein teen baar diya. CBC test karwana hai.”

Same note in Devanagari: “मरीज़ को 3 दिन से तेज़ बुखार है, सिर दर्द भी है। Paracetamol 650 mg दिन में तीन बार दिया। CBC test करवाना है।”

What MediBridge returns

Hinglish span	English span	Type
tez bukhar	high fever	CLINICAL_FINDING
sar dard	headache	CLINICAL_FINDING
Paracetamol 650 mg	Paracetamol 650 mg	DRUG
din mein teen baar	three times a day	ADVICE
CBC test	complete blood count test	DIAGNOSTIC

+ readable English: “The patient has had a high fever for 3 days, with a headache. Gave Paracetamol 650 mg three times a day. CBC test advised.”

English-only EHR / ABDM records can’t parse it

A search for “fever” misses “bukhar”

Claim fields stay blank → staff re-type by hand

Illustrative note written by the team to show the problem; not real patient data.

Problem brief: the scale of the gap

52.8 crore

people speak Hindi (incl. dialects) — 43.6% of India

Census 2011

13.9 lakh

registered allopathic doctors; doctor–population ratio ≈ 1:811

*Rajya Sabha reply, Dec 2025**

~2 hours

of EHR & desk work for every 1 hour with patients

Sinsky et al., Ann. Intern. Med. 2016

104 crore

health records now linked to ABHA IDs — records are going digital

PIB / ABDM, Jul 2026

The problem

- Doctors and patients in Hindi-speaking regions talk and write in Hinglish — Hindi grammar with English medical terms, in Roman or Devanagari script.
- The systems downstream — hospital EHRs, ABDM health records, medical coding and insurance claims — expect structured English.
- Today, that gap is closed by hand: staff translate and re-type notes. It costs time, adds errors, and makes Hinglish records unsearchable.

**Ratio counts allopathic + AYUSH practitioners and assumes 80% availability. Hindi figure includes dialects (Hindi alone ≈ 32.2 crore). Full references on the last slide.*

Problem statement and scope

Given a Hinglish clinical note (typed, or transcribed from speech), produce a readable English version and a list of medical entities — each with its original Hindi/Hinglish span, its English span, and an English entity type — so the note can be used by English-only health systems.

In scope

- Hindi–English code-mixed text (Roman and Devanagari)
- Hinglish → English translation with span alignment
- Entity extraction into 5 English types (below)
- Confidence score per entity + human review flag
- Speech input via an ASR front-end (built in M5)

Out of scope

- Diagnosis, treatment advice or any clinical decision
- Languages other than Hindi and English
- Real patient records (public, licensed data only)
- Full ontology coding (SNOMED / ICD) — a stretch goal
- Handwriting / scanned-document OCR

Entity types:

DRUG	CLINICAL_FINDING	DIAGNOSTIC	ADVICE	MISC_MEDICAL
medicine + dose	symptom / disease	test / exam	frequency, diet, rest	other medical terms

Entity types follow the clinician-annotated scheme of our chosen dataset (Eka Medical ASR Evaluation Dataset, analysed in M2).

User identification: who uses MediBridge

PRIMARY

OPD doctor

Who: Clinic or hospital physician who writes or dictates notes in Hinglish.

Pain: Hours of documentation after clinic; notes unreadable to English systems.

Needs: One-click English note + structured fields, with the Hindi span kept so they can verify.

Example: *Types "BP high hai, Amlodipine 5 mg OD shuru" → "BP high; started Amlodipine 5 mg once daily".*

PRIMARY

Records staff / medical coder

Who: Transfers notes into the EHR, ABDM records and coding systems.

Pain: Manually translates and re-types every Hinglish note; slow and error-prone.

Needs: An entity table they only review, with low-confidence items flagged first.

Example: *Sees DRUG = Amlodipine, DOSAGE = 5 mg once daily, pre-filled; only checks flagged rows.*

SECONDARY

Insurer / TPA claims processor

Who: Reviews hospital documents to approve health-insurance claims.

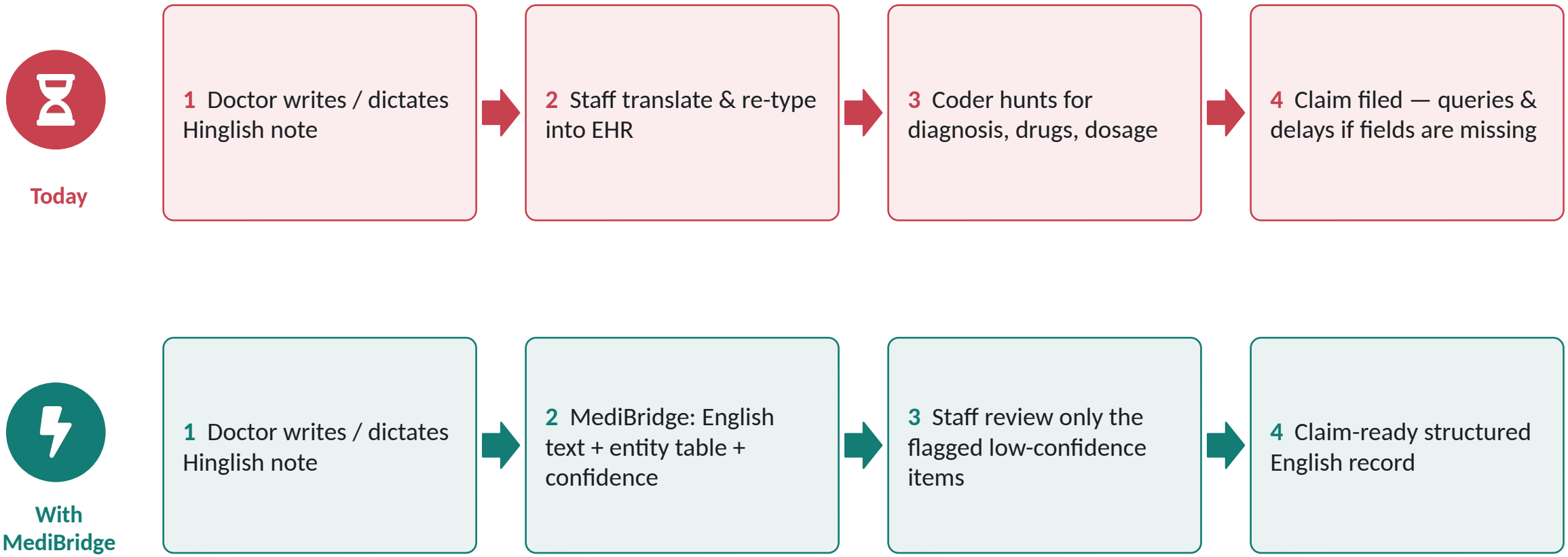
Pain: Incomplete or unreadable notes cause queries, delays and disputes.

Needs: Standard English diagnosis, drug, dosage, duration and procedure fields.

Example: *Receives DISEASE = hypertension with the source Hindi span attached as evidence.*

Also affected: patients (records they and the next doctor can read, portable through ABHA) · **Buyer:** hospital / clinic-chain administration and TPAs, who pay for documentation time and claim delays.

User journey: today vs. with MediBridge



The doctor's step stays the same — MediBridge removes the manual translation and re-typing in the middle, and turns full review into exception review.

Business objective

What we want to achieve

O1 — Cut documentation effort
Remove manual translation and re-typing of Hinglish notes for doctors and records staff.

O2 — Produce claim-ready data
Turn every note into standard English fields (diagnosis, drug, dosage, duration, procedure).

O3 — Make Hinglish records usable
Searchable, readable English records that fit EHR / ABDM systems.

Business model: B2B SaaS per clinic / per-note API for hospitals, clinic chains and TPAs. Runs on-premise with open models — no patient data leaves the hospital.

Evidence the value is real

≈15,800 hrs

of documentation saved by AI scribes at Kaiser Permanente: 7,260 physicians, ~2.5M encounters, Oct 2023–Dec 2024 (self-reported estimate).

3,000+ / 4,000+

Indian doctors using Eka Care's EkaScribe (Hinglish-capable clinical ASR) / Augnito medical speech AI (vendor-reported).

₹26,000 cr

of health-insurance claims repudiated in FY24, up 19.1% (IRDAI). Documentation gaps are one cited cause — not the only one.

Sources: TPMG / NEJM Catalyst 2025 via AMA & Becker's; Eka Care & Augnito press coverage; IRDAI Annual Report FY24 via Business Standard.

Proposed solution: alignment-aware pipeline

Input

Typed note, or speech → Whisper ASR

1 Normalise

Unicode + nukta, spelling; spot English written in Devanagari

2 Translate + align

IndicTrans2 → English; awesome-align word map

3 English NER

GLiNER, zero-shot, 5 clinical types

4 Project back

English entity → original Hindi span

5 Confidence

C = C_trans × C_align × C_NER; low → human review

Output record (one entity)

```
{
  "hindi_span": "tez bukhar",
  "english_span": "high fever",
  "canonical": "fever",
  "type": "CLINICAL_FINDING",
  "src_offsets": [20, 30],
  "tgt_offsets": [22, 32],
  "confidence": 0.85,
  "needs_review": false
}
```

- Primary: translate + align → English NER → project (keeps Hindi ↔ English offsets).
- Validation path: detect Hindi spans → translate only the spans → type.
- English terms are kept, incl. those written in Devanagari (डायबिटीज = diabetes).

Success metrics

Tier	Metric	Target	Why this target
Model	Entity F1 vs clinician gold, relaxed (micro)	**≥ 0.70**	Scored on our leak-free hi test split: 47 transcripts, 285 gold spans
Model	Entity F1, strict (exact span + type)	**≥ 0.60**	Exact boundaries are harder after translation (Jain et al. 2019)
Model	Macro-F1 over the 5 types	**≥ 0.55**	Types are imbalanced about 14 : 1 (M2), so rare types must count
Model	F1 gap: conversation vs narration	**≤ 0.10**	Type mix depends on context (Cramér's V = 0.54, M2)
Model	English-span accuracy (100 entities, manual)	**≥ 85%**	Gold labels are Hindi-side; 2 reviewers check the English
Model	Speech WER on Hindi medical audio (M5)	**≤ 20%**	Whisper-v3 15.7% on Eka English set; IndicWhisper 13.6% Hindi
System	p95 latency per note (≤ 100 words, CPU)	**≤ 2 s**	Must feel instant inside an OPD workflow
System	Cost per note	**≈ ₹0 API cost**	All models open-source and run locally
Business	Manual entry time per note (timed test)	**−50%**	Team-timed study: manual vs. review-only, 10 notes
Business	Claim fields auto-filled when present	**≥ 90%**	Drug, finding, diagnostic and advice fields

Targets are our proposals for M4, grounded in the M2 data analysis and the cited baselines. Gold labels: clinician-annotated Hindi-side spans.

Related work and past experiments we build on

Work	Task / data	Reported result	What we reuse
Singh et al., 2018 (ACL NEWS workshop)	NER on Hindi–English code-mixed tweets	F1 0.95 with CRF / LSTM	**Shows code-mixed NER is learnable; tokenisation ideas**
Jain et al., 2019 (EMNLP)	Entity projection via machine translation	+4.1 F1 avg. over prior SOTA, 5 languages	**Our translate-and-project design**
IndicTrans2 — Gala et al., 2023 (TMLR)	MT for all 22 scheduled Indian languages	Open, MIT licence; 200M distilled hi→en	**Translation engine**
GLiNER — Zaratiana et al., 2024 (NAACL)	Zero-shot NER with any label set	Beats ChatGPT zero-shot; runs on CPU	**English NER with our 5 types**
awesome-align — Dou & Neubig, 2021 (EACL)	Word alignment from mBERT	Beat prior aligners on 5 language pairs	**Hindi ↔ English span alignment**
MedSumm — Ghosh et al., 2024 (ECIR)	Summarising Hinglish medical queries	3,015 queries with English gold summaries	**Data + 'readable English' evaluation**
Vistaar / IndicWhisper, 2023 (Interspeech)	ASR benchmark for Indian languages	Hindi avg. WER 13.6%	**Speech front-end (M5)**
Eka Medical ASR Eval — EkaCare, 2025	Indian medical speech + clinician entity spans	Whisper-v3 WER 15.7%; best 10.9% (English)	**Our dataset (M2) + speech baseline**

Results as reported by the authors. Full citations and links on the References slide.

Feasibility: data (chosen and prepared in M2)

Primary dataset: Eka Medical ASR Evaluation Dataset (EkaCare, 2025, MIT licence): doctor–patient Hinglish speech with transcripts and clinician-annotated entity spans.

316

clean Hinglish transcripts (1.57 h of audio, 203 sessions)

1,733

clinician-labelled entity spans, plus 10,309 in the English subset

221 / 48 / 47

train / val / test transcripts, zero session or text overlap

0

phone, e-mail or ID numbers found; 2 names masked as [NAME]

Supporting dataset	What it contains	Licence	Role in project
PHINC (WNUT 2020)	13,738 Hinglish → English sentence pairs	CC-BY-4.0	Evaluating the translation step
MMCQS (MedSumm, ECIR 2024)	3,015 Hinglish medical queries + English summaries	Released by authors	Extra Hinglish medical text
COMI-LINGUA (2025)	125K+ code-mixed instances incl. NER	CC-BY-4.0	General code-mixed NER check

Remaining gap: gold labels are Hindi-side spans only. The English spans are our contribution: 2 reviewers check 100 of them (Cohen's κ). The Hindi test set is small, so we report bootstrap confidence intervals.

What the data told us: M2 findings that shape the design

93%

of Hindi entity spans are written in Devanagari, even English words: डायबिटीज = diabetes, एमजी = mg.

So we: The normaliser must recognise English written in Devanagari before translation.

68–72%

of entity mentions in the val/test splits never appear in training.

So we: A dictionary would fail. Use zero-shot NER (GLiNER) with translation.

V = 0.54

Entity type depends strongly on context: conversations carry the tests and advice, narrations the drugs.

So we: Splits are stratified by context; metrics are reported per context.

14 : 1

class imbalance between the most frequent type (drugs) and the rarest (misc_medical).

So we: Report macro-F1 and use class weights when fine-tuning.

Source: our M2 analysis of the Eka Medical ASR Evaluation Dataset, Hindi subset (χ^2 test, Cramér's V, out-of-vocabulary rates). Notebooks in m2_data/.

Feasibility: technology, compute and team

Models (all open-source)

- IndicTrans2 hi→en, distilled: 200M params (MIT)
- GLiNER: small / medium / large variants, CPU-friendly (Apache-2.0)
- awesome-align on multilingual BERT-base
- Whisper small (244M) / IndicWhisper for speech

Compute

- Inference: every model runs on a laptop CPU
- Fine-tuning (M3–M4): team GPU access
- No paid APIs → no per-note cost, no data leaves the machine
- Stack: Python, PyTorch, Hugging Face, FastAPI / Streamlit, Git

Team of 4: roles

- Data lead (M2 pipeline, English-span review)
- Translation & alignment (IndicTrans2, awesome-align)
- NER, normalisation & evaluation (GLiNER, metrics)
- App, deployment & reporting (API, demo, docs)

Verdict: feasible. Every component is open-source with a permissive licence and runs locally. Labelled data is in hand (M2); the main risk is the small Hindi set.

Risks, mitigations and privacy

Risk	Impact	Mitigation
English written in Devanagari (93% of entity spans, M2)	Missed or mistranslated entities	Transliteration-aware normalisation; keep English terms as-is
Word order changes break alignment	Wrong Hindi span returned	Alignment confidence; fall back to span-only translation path
Small Hindi set (316 transcripts, 5 known speakers)	Unstable estimates	Leak-free splits; English subset as auxiliary; bootstrap CIs
Unseen entities (68–72% of test mentions, M2)	Memorising models fail	Zero-shot GLiNER + translation; subword models
Wrong drug or dose extracted	Clinical-safety risk	Human review below threshold; tool never gives medical advice
Staged conversations, noisy real audio	Lower accuracy in clinics	Report per context; speech in M5 with ASR confidence flag



Privacy by design

- India's DPDP Act 2023; DPDP Rules notified Nov 2025, phasing in through 2026–27
- M2 scan: 0 phone, e-mail or ID hits; 2 names masked as [NAME]
- Audio never redistributed; raw data kept out of git
- On-premise models: notes never go to a third-party API

Roadmap: M1 to M6



Problem & business case

This deck: problem, users, objective, metrics, feasibility

Data preparation

Done: Eka dataset cleaned, analysed, leak-free splits, privacy

Baseline & models

Translate → GLiNER baseline; aligned pipeline; experiment log

Evaluation & optimisation

F1 vs clinician gold, per context; error analysis; fine-tuning

Deployment prototype

FastAPI + web UI, speech input, monitoring, failure handling

Final delivery

Technical report, GitHub repo, live demo, business pitch

Next (M3): baseline on the M2 splits: translate → GLiNER → project back to Hindi, scored against the clinician gold spans, with every run logged in Git.

References

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[4] Ministry of Health & FW, Rajya Sabha written reply, Dec 2025 (doctor–population ratio 1:811)

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